

MATHEMATICS-III

Unit-I: Differential Equations

Comprehensive Study Notes as per HPU BCA Syllabus

Topic 1: Order and Degree of Differential Equations

1.1 What is a Differential Equation?

A **differential equation** is an equation that contains one or more **derivatives** of an unknown function.

General Form:

$$F(x, y, y', y'', \dots, y^n) = 0$$

Examples:

$$dy/dx = x + y$$

$$d^2y/dx^2 + 3dy/dx + 2y = 0$$

1.2 Order of a Differential Equation

The **order** of a differential equation is the **highest order derivative** present in the equation.

Differential Equation	Order
$dy/dx = x^2$	1
$d^2y/dx^2 + 2dy/dx + y = 0$	2
$d^3y/dx^3 + 3x^2dy/dx = \sin x$	3

1.3 Degree of a Differential Equation

The **degree** of a differential equation is the **highest power** of the highest order derivative (provided the equation is polynomial in derivatives).

Steps to Find Degree:

1. The equation must be free from radicals and fractions.
2. The power of the highest order derivative is the degree.

Differential Equation	Degree
$(dy/dx)^2 + 2y = 0$	2
$(d^2y/dx^2)^3 + dy/dx = 0$	3
$(dy/dx)^{1/2} + x = 0$	Not polynomial

1.4 Examples: Finding Order and Degree

Example 1: Find order and degree of $d^2y/dx^2 + 3dy/dx + 2y = 0$

Feature	Value
Highest derivative	d^2y/dx^2
Order	2
Power of highest derivative	1
Degree	1

Example 2: Find order and degree of $(dy/dx)^2 = \sin x$

Feature	Value
Highest derivative	dy/dx
Order	1
Power of highest derivative	2

Degree	2
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Example 3: Find order and degree of $(d^2y/dx^2)^3 + 2(dy/dx)^2 + y = 0$

Feature	Value
Highest derivative	d^2y/dx^2
Order	2
Power of highest derivative	3
Degree	3

Topic 2: Solution of Differential Equations

2.1 Types of Solutions

Type	Description
General Solution	Contains arbitrary constants (as many as the order)
Particular Solution	Obtained by giving specific values to constants
Singular Solution	Not contained in the general solution

2.2 Variable Separable Method

If equation is of the form:

$$dy/dx = f(x) \cdot g(y)$$

Steps:

1. Separate variables: $dy/g(y) = f(x) dx$
2. Integrate both sides
3. Add constant of integration

Example: Solve $dy/dx = 2xy$

$$\begin{aligned} dy/y &= 2x dx \\ \int dy/y &= \int 2x dx \\ \ln|y| &= x^2 + C \\ y &= C_1 e^{x^2} \end{aligned}$$

2.3 Homogeneous Differential Equations

If the equation is of the form:

$$dy/dx = f(x,y) / g(x,y)$$

where f and g are homogeneous functions of the same degree.

Substitution: Let $y = vx$, then $dy/dx = v + x \cdot dv/dx$

Example: Solve $dy/dx = (x + y)/(x - y)$

$$\text{Let } y = vx \Rightarrow dy/dx = v + x \cdot dv/dx$$

$$v + x \cdot dv/dx = (x+vx)/(x-vx) = (1+v)/(1-v)$$

$$x \cdot dv/dx = (1+v)/(1-v) - v$$

$$x \cdot dv/dx = (1+v - v + v^2)/(1-v)$$

$$x \cdot dv/dx = (1+v^2)/(1-v)$$

$$(1-v)/(1+v^2) \cdot dv = dx/x$$

Integrating gives the solution.

2.4 Linear Differential Equations

If the equation is of the form:

$$dy/dx + P(x) \cdot y = Q(x)$$

Solution Steps:

1. Find integrating factor: $IF = e^{\int P(x) dx}$
2. Solution: $y \cdot IF = \int Q(x) \cdot IF dx + C$

Example: Solve $dy/dx + 2y = e^x$

$$P(x) = 2, Q(x) = e^x$$

$$IF = e^{\int 2 dx} = e^{2x}$$

$$y \cdot e^{2x} = \int e^x \cdot e^{2x} dx = \int e^{3x} dx$$

$$y \cdot e^{2x} = (1/3)e^{3x} + C$$

$$y = (1/3)e^x + C \cdot e^{-2x}$$

Topic 3: Formation of Differential Equations

3.1 Steps to Form Differential Equation

1. Write the equation of the curve/function with constants.
2. Differentiate as many times as the number of constants.
3. Eliminate the constants.

Example 1: Form differential equation of $y = Ae^{2x} + Be^{-3x}$

$$\begin{aligned}y &= Ae^{2x} + Be^{-3x} \\y' &= 2Ae^{2x} - 3Be^{-3x} \\y'' &= 4Ae^{2x} + 9Be^{-3x}\end{aligned}$$

Eliminating A and B:

$$y'' - y' - 6y = 0$$

Example 2: Form differential equation of circle $x^2 + y^2 = r^2$

$$\begin{aligned}x^2 + y^2 &= r^2 \\2x + 2y \cdot dy/dx &= 0 \\x + y \cdot dy/dx &= 0\end{aligned}$$

Topic 4: Linear Differential Equations with Constant Coefficients

4.1 Standard Form

$$(D^n + a_1 D^{n-1} + \dots + a_n)y = f(x)$$

where $D = d/dx$, $D^2 = d^2/dx^2$, etc.

4.2 Solution of Homogeneous Equation $f(D)y = 0$

Steps:

1. Write auxiliary equation: $m^n + a_1 m^{n-1} + \dots + a_n = 0$
2. Find roots ($m_1, m_2, \dots, m_n$)
3. Write complementary function:

Roots

Complementary Function (C.F.)

Real and distinct	$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + \dots$
Real and equal	$y = (C_1 + C_2 x + \dots) e^{mx}$
Complex $\alpha \pm i\beta$	$y = e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$

4.3 Particular Integral for $f(D)y = Q(x)$

$Q(x)$

Particular Integral (P.I.)

e^{ax}	$e^{ax} / f(a)$ if $f(a) \neq 0$
$x e^{ax}$	$x e^{ax} / f'(a)$ if $f(a) = 0$

$\sin(ax)$ or $\cos(ax)$	$\sin(ax)/f(-a^2)$ or $\cos(ax)/f(-a^2)$
x^n	$1/f(D) \cdot x^n$ (using expansion)

4.4 General Solution

$$y = \text{C.F.} + \text{P.I.}$$

Example 1: Solve $(D^2 + 3D + 2)y = 0$

Auxiliary equation:

$$m^2 + 3m + 2 = 0$$

$$(m+1)(m+2) = 0$$

$$m = -1, -2$$

Complementary Function:

$$y = C_1 e^{-x} + C_2 e^{-2x}$$

Example 2: Solve $(D^2 - 4D + 4)y = e^{2x}$

Auxiliary equation:

$$m^2 - 4m + 4 = 0$$

$$(m-2)^2 = 0$$

$$m = 2, 2$$

$$\text{C.F.} = (C_1 + C_2 x)e^{2x}$$

$$\text{P.I.} = 1/(D^2 - 4D + 4) \cdot e^{2x}$$

Since $f(2) = 0$, use $x \cdot e^{2x}/f'(2)$

$$f'(D) = 2D - 4, \quad f'(2) = 4 - 4 = 0, \quad \text{so use } x^2 \cdot e^{2x}/f''(2)$$

$$f''(D) = 2, \quad f''(2) = 2$$

$$\text{P.I.} = x^2 e^{2x}/2$$

General Solution:

$$y = (C_1 + C_2 x)e^{2x} + x^2 e^{2x}/2$$

Topic 5: Cauchy's Homogeneous Equation

5.1 Standard Form

$$x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_n y = f(x)$$

5.2 Solution Steps

1. Put $z = \log x$ or $x = e^z$.
2. Let $x \frac{d}{dx} = \theta$, $x^2 \frac{d^2}{dx^2} = \theta(\theta-1)$, ...
3. Solve as linear equation in θ .

Example: Solve $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = x^2$

Let $x = e^z$, $x \frac{dy}{dx} = \theta y$, $x^2 \frac{d^2 y}{dx^2} = \theta(\theta-1)y$

Equation becomes:

$$[\theta(\theta-1) + \theta - 1]y = e^{2z}$$

$$(\theta^2 - 1)y = e^{2z}$$

Auxiliary equation:

$$m^2 - 1 = 0 \Rightarrow m = \pm 1$$

$$\text{C.F.} = C_1 e^z + C_2 e^{-z} = C_1 x + C_2/x$$

$$\text{P.I.} = \frac{1}{(\theta^2 - 1)} \cdot e^{2z} = \frac{e^{2z}}{3} = \frac{x^2}{3}$$

General Solution:

$$y = C_1 x + C_2/x + x^2/3$$

Topic 6: Legendre's Equation

6.1 Standard Form

$$(ax+b)^n \frac{d^n y}{dx^n} + a_1(ax+b)^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_n y = f(x)$$

6.2 Solution Steps

1. Put $z = \log(ax+b)$.
2. Let $(ax+b) \frac{d}{dx} = \theta$.
3. Solve as linear equation in θ .

Example: Solve $(x+1)^2 \frac{d^2 y}{dx^2} + (x+1) \frac{dy}{dx} + y = 0$

Let $z = \log(x+1)$, $\theta = (x+1) \frac{d}{dx}$

Equation becomes:

$$[\theta(\theta-1) + \theta + 1]y = 0$$

$$(\theta^2 + 1)y = 0$$

Auxiliary equation:

$$m^2 + 1 = 0 \Rightarrow m = \pm i$$

$$\text{C.F.} = C_1 \cos z + C_2 \sin z$$

$$y = C_1 \cos[\log(x+1)] + C_2 \sin[\log(x+1)]$$

Topic 7: HPU Important Questions

Short Answer Questions (2–5 Marks)

1. What is a differential equation?
2. Define order and degree of a differential equation.
3. What is the general solution of a differential equation?
4. What is the complementary function?
5. What is a particular integral?
6. Write the auxiliary equation for $(D^2 + 3D + 2)y = 0$.
7. What is Cauchy's homogeneous equation?
8. What is Legendre's equation?
9. What is variable separable method?
10. What is a linear differential equation?

Long Answer Questions (10 Marks)

1. Solve $dy/dx + y = x^2$.
2. Solve $(D^2 + 4D + 4)y = e^{-2x}$.
3. Solve $x^2 d^2y/dx^2 + x dy/dx - 4y = 0$.
4. Solve $(2x+3)^2 d^2y/dx^2 + (2x+3) dy/dx - y = x$.
5. Form the differential equation for $y = Ae^{3x} + Be^{-2x}$.
6. Solve $dy/dx = (x^2 + y^2)/(xy)$.

Objective Questions (1 Mark)

1. Order of $d^2y/dx^2 + 3dy/dx + 2y = 0$ is:
(a) 1 (b) 2 (c) 3 (d) 4 **Answer: (b)**
2. Degree of $(dy/dx)^3 = 0$ is:
(a) 1 (b) 2 (c) 3 (d) 0 **Answer: (c)**
3. Integrating factor for $dy/dx + 2y = x$ is:
(a) e^{2x} (b) e^x (c) e^{-2x} (d) 1 **Answer: (a)**

4. Auxiliary equation for $(D^2 + 4)y = 0$ is:

(a) $m^2 + 4 = 0$ (b) $m^2 - 4 = 0$ (c) $m + 4 = 0$ (d) $m - 4 = 0$ **Answer: (a)**

Quick Revision

Order and Degree

Term	Definition
Order	Highest derivative
Degree	Highest power of highest derivative

Linear Differential Equation

$$dy/dx + P(x)y = Q(x) \Rightarrow y \cdot IF = \int Q(x) \cdot IF \, dx + C, \text{ where } IF = e^{\int P(x) \, dx}$$

D-Operator Method

Term	Meaning
D	d/dx
D^2	d^2/dx^2
$f(D)y = 0$	Homogeneous equation
$y = \text{C.F.} + \text{P.I.}$	General solution

Auxiliary Equation Roots → C.F.

Roots	C.F.
m_1, m_2 (real, distinct)	$C_1 e^{m_1 x} + C_2 e^{m_2 x}$

$m, m \text{ (equal)}$	$(C_1 + C_2x)e^{mx}$
$\alpha \pm i\beta$	$e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$

End of Unit-I Notes — MATHEMATICS-III (HPU BCA)